

Independent Replication Report: *Is Grover's Algorithm a Quantum Hidden Subgroup Algorithm?*

Lomonaco & Kauffman, arXiv:quant-ph/0603140 (2006)

QC-200 replication wave — automated agent report

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Abstract

This is an independent numerical replication of Lomonaco and Kauffman (arXiv:quant-ph/0603140). The paper is largely theoretical: it recasts Grover's search algorithm as a non-abelian Hidden Subgroup Problem (HSP) on the symmetric group S_N with hidden subgroup Stab_{j_0} , and shows via two group-theoretic arguments that the *standard* non-abelian QHS algorithm cannot solve this particular HSP. We reproduce every quantitative and structural claim that can be reduced to a finite computation: (A) exact Grover success probability for $N \in \{4, 8, 16\}$ via Qiskit statevector simulation (matches the paper's closed-form $P_k = \sin^2((2k+1)\theta)$ to $\sim 10^{-14}$); (B) invariance of the Grover state at optimal k under every element of $\text{Stab}_{j_0} \subset S_N$ (fidelity = 1, verified for 20 random elements) and non-invariance under permutations outside Stab_{j_0} ; (C) Proposition 1 of Section 7 (the N transpositions $\{(0\ j_0), \dots, ((N-1)\ j_0)\}$ form a complete transversal of S_N/Stab_{j_0}), verified by explicit coset enumeration for $N \in \{3, 4, 5\}$; (D) both parts of Section 9's "However" (largest normal subgroup of S_N contained in Stab_{j_0} is trivial; the stabilisers $\text{Stab}_0, \dots, \text{Stab}_{N-1}$ are mutually conjugate); (F) the Section 8 pushed-oracle $\tilde{\phi} = \phi \circ \tau$ is information-theoretically identical to Grover's Boolean oracle $f(j) = [j = j_0]$. **Verdict: REPLICATED.**

1 Paper summary

Lomonaco and Kauffman investigate whether Grover's search algorithm belongs to the same conceptual family as Shor's algorithm — the family of *quantum hidden subgroup* (QHS) algorithms. They give two answers:

1. Grover's algorithm *can* be formulated as a QHS-style algorithm. Let S_N act on the database labels $\{0, 1, \dots, N-1\}$ by permutation. The stabiliser

$$\text{Stab}_{j_0} = \{g \in S_N : g(j_0) = j_0\}$$

of the marked element j_0 is a subgroup of S_N , and Grover's evolution commutes with the group action of Stab_{j_0} on the database Hilbert space $H = \mathbb{C}^N$. Knowing Stab_{j_0} is equivalent to knowing j_0 . Section 7 gives the diagram

$$\begin{array}{ccc} S_N & \xrightarrow{\phi} & S = \{0, \dots, N-1\} \\ & \searrow \nu_{j_0} & \nearrow \iota \\ & S_N/\text{Stab}_{j_0} & \end{array}$$

and the paper's Section 8 shows that Grover's Boolean oracle $f(j) = [j = j_0]$ arises as $\tilde{\phi} = \phi \circ \tau$ with the transversal $\tau : (0\ j)\text{Stab}_0 \mapsto (0\ j)$.

2. However (Section 9), the *standard* non-abelian QHS algorithm on S_N *cannot* solve this HSP, for two group-theoretic reasons:
 - (i) The Hallgren–Russell–Ta-Shma theorem [?] implies that Fourier-sampling on S_N can only reveal the largest *normal* subgroup of S_N contained in Stab_{j_0} ; that largest normal subgroup is $\{e\}$ (trivial).
 - (ii) $\text{Stab}_0, \dots, \text{Stab}_{N-1}$ are pairwise conjugate in S_N , hence indistinguishable to any conjugation-invariant statistic.

The paper leaves open whether a *modified* QHS algorithm could solve Grover’s HSP. Zalka’s optimality result [?] implies that any such algorithm cannot beat $O(\sqrt{N})$ queries.

2 Claims table

ID	Claim (as stated / implied by paper)	Testable?	Tested
C1	Standard Grover: $P_k(j_0) = \sin^2((2k+1)\theta)$, $\theta = \arcsin(1/\sqrt{N})$	yes (statevector)	yes
C2	Optimal $k = \lfloor \pi/(4\theta) \rfloor$ gives $P \geq 1 - 1/N$	yes	yes
C3	Grover state is invariant under Stab_{j_0} action	yes (unitary $\otimes$ state)	yes
C4	Grover state is NOT invariant under generic $g \notin \text{Stab}_{j_0}$	yes	yes
C5	Prop. 1: $\{(k, j_0) : k = 0..N-1\} = \text{full transversal of } S_N/\text{Stab}_{j_0}$	yes (finite)	yes
C6	(Sec. 9-i) Largest normal subgroup of S_N in Stab_{j_0} is $\{e\}$	yes (finite)	yes
C7	(Sec. 9-ii) $\text{Stab}_0, \dots, \text{Stab}_{N-1}$ mutually conjugate	yes (finite)	yes
C8	Section-8 pushed oracle $\tilde{\phi}$ is Grover’s Boolean oracle f	yes (finite)	yes
C9	Standard non-abelian QHS on S_N cannot identify j_0 from Fourier samples	yes (argument via C6/C7)	yes (in)
C10	No QHS-Grover variant can beat $O(\sqrt{N})$ (Zalka bound)	no (asymptotic)	no

Table 1: Claims and whether they were tested. C1–C8 are directly reproducible. C9 is confirmed through the C6+C7 argument (Hallgren–Russell–Ta-Shma reduces Fourier-sampling’s information content to normal-subgroup detection; we verify that normal-content is trivial). C10 is an asymptotic-optimality claim from [?] and is outside the scope of a finite numerical check.

3 Method

3.1 Software stack

- Python 3.14 (system python3), venv `~/qc_venv`
- qiskit 2.5.0, numpy 2.5.1, sympy 1.14.0
- Simulator: `qiskit.quantum_info.Statevector` (exact statevector, no shots)

3.2 Numbered procedure (exact commands)

1. Fetch paper: `curl -sL -o work/0603140.pdf https://arxiv.org/pdf/quant-ph/0603140`.
2. `pdftotext` the PDF into `work/0603140.txt` for reading.
3. Set up venv: `python3 -m venv ~/qc_venv && source ~/qc_venv/bin/activate && pip install qiskit qiskit-aer numpy sympy`.
4. Run `code/grover_hsp.py` which executes Parts A–F below.

3.3 Part A — Standard Grover, statevector, $N \in \{4, 8, 16\}$

For $n \in \{2, 3, 4\}$, $N = 2^n$, marked element $j_0 = 1$: build the textbook Grover circuit (initial Hadamards, then k repetitions of oracle $I - 2|j_0\rangle\langle j_0|$ followed by diffuser $2|s\rangle\langle s| - I$) and simulate exactly with `Statevector.from_instruction`. Compare $|\langle j_0 | \psi_k \rangle|^2$ to $\sin^2((2k+1) \arcsin(1/\sqrt{N}))$ for $k = 0, \dots, 2k_{\text{opt}} + 2$.

3.4 Part B — Invariance under Stab_{j_0}

For each $N \in \{4, 8, 16\}$, run Grover to the optimal k , obtain the N -dimensional statevector $|\psi\rangle$. Draw 20 random permutations of $\{0, \dots, N-1\}$ fixing j_0 (elements of Stab_{j_0}) and another 20 random permutations *not* fixing j_0 . Build the $N \times N$ permutation unitary U_g and compute the fidelity $|\langle \psi | U_g \psi \rangle|^2$.

3.5 Part C — Coset structure (Prop. 1)

For $N \in \{3, 4, 5\}$, enumerate all $N!$ permutations, form Stab_{j_0} $((N-1)!$ elements), form the N candidate coset representatives $(k j_0)$, generate the N left cosets $(k j_0) \cdot \text{Stab}_{j_0}$, and check that they are (a) pairwise disjoint, (b) each of size $(N-1)!$, (c) their union equals S_N .

3.6 Part D — Section 9 (“However”)

For $N \in \{3, 4, 5\}$: enumerate the normal subgroups of S_N we can name in closed form ($\{e\}$, V_4 for $N = 4$, A_N , S_N) and check which of them are contained in Stab_0 . Also verify that $(i j) \cdot \text{Stab}_i \cdot (i j)^{-1} = \text{Stab}_j$ for every $i \neq j$.

3.7 Part E — QHS indistinguishability

Report the induced-representation dimensions and known irreducible decomposition of $\text{Ind}_{\text{Stab}_0}^{S_N}(\mathbf{1})$ vs $\text{Ind}_{\{e\}}^{S_N}(\mathbf{1})$. Combined with the Hallgren–Russell–Ta-Shma theorem (which *cites*, not *proves*), this gives the paper’s Section-9 conclusion.

3.8 Part F — Pushed-oracle equivalence

For $N \in \{3, 4, 5\}$ and $j_0 \in \{2, 1, 3\}$: for each $j \in \{0, \dots, N-1\}$, compute the coset $(0 j) \cdot \text{Stab}_{j_0}$ and check whether it equals $(0 j_0) \cdot \text{Stab}_{j_0}$; this evaluates the pushed oracle to 1 or 0, which must agree with $f(j) = [j = j_0]$.

4 Results vs paper

4.1 Part A: Grover success probability (headline)

N	$\theta = \arcsin(1/\sqrt{N})$	k_{opt}	$P_{\text{sim}}(k_{\text{opt}})$	$P_{\text{formula}}(k_{\text{opt}})$	$ \text{err} $
4	0.523599	1	1.0000000000	1.0000000000	1.6×10^{-15}
8	0.361367	2	0.9453125000	0.9453125000	4.2×10^{-15}
16	0.252680	3	0.9613189697	0.9613189697	8.7×10^{-15}

Table 2: Statevector-simulated Grover success probability at optimal k vs the paper’s closed-form formula $\sin^2((2k+1)\theta)$. Agreement is to floating-point precision. All three exceed the paper’s stated bound $1 - 1/N$. **C1, C2 REPRODUCED.**

The full k -curve (Part A includes $k = 0..2k_{\text{opt}} + 2$) is in `report/evidence/results.json` and shows the expected sinusoidal overshoot–undershoot pattern.

4.2 Part B: Stabiliser invariance

N	min fid over Stab	mean fid over Stab	max fid over non-Stab	# non-Stab with fid < 0.999
4	1.0	1.0	0.0000	20/20
8	1.0	1.0	0.0156	20/20
16	1.0	1.0	0.0040	20/20

Table 3: For 20 random permutations in Stab_{j_0} vs 20 outside: Grover state is invariant to machine precision under Stab_{j_0} and clearly moved (fidelity near zero) by any generic non-stabiliser permutation. **C3, C4 REPRODUCED.**

4.3 Part C: Coset transversal (Prop. 1)

- $N = 3$, $j_0 = 1$: 3 cosets, sizes $[2, 2, 2]$, disjoint, cover S_3 ($|S_3| = 6$). **OK**
- $N = 4$, $j_0 = 2$: 4 cosets, sizes $[6, 6, 6, 6]$, disjoint, cover S_4 ($|S_4| = 24$). **OK**
- $N = 5$, $j_0 = 0$: 5 cosets, sizes $[24, 24, 24, 24, 24]$, disjoint, cover S_5 ($|S_5| = 120$). **OK**

C5 REPRODUCED.

4.4 Part D: “However” section

- $N = 3$: $\{e, A_3, S_3\}$ enumerated; only $\{e\}$ contained in Stab_0 , so largest-normal-in-stab is trivial. All three stabilisers mutually conjugate.
- $N = 4$: $\{e, V_4, A_4, S_4\}$ enumerated; $V_4 = \{e, (01)(23), (02)(13), (03)(12)\}$ — every non-identity element moves index 0, so $V_4 \cap \text{Stab}_0 = \{e\}$; likewise A_4 and S_4 contain elements moving 0. Largest-normal-in-stab is trivial. All four stabilisers mutually conjugate.
- $N = 5$: $\{e, A_5, S_5\}$ (A_5 is simple); largest-normal-in-stab trivial. All five stabilisers mutually conjugate.

C6, C7 REPRODUCED.

4.5 Part E: QHS indistinguishability (structural)

For every N , $\dim \text{Ind}_{\text{Stab}_0}^{S_N}(\mathbf{1}) = N$ (the trivial+standard permutation representation on N points), while $\dim \text{Ind}_{\{e\}}^{S_N}(\mathbf{1}) = N!$ (the regular representation). Combined with the Hallgren–Russell–Ta-Shma structural theorem, C9 follows. This confirms the paper’s conclusion without our having to re-derive HRT.

4.6 Part F: Pushed oracle = Grover oracle

For $(N, j_0) \in \{(3, 2), (4, 1), (5, 3)\}$: iterated $j = 0, \dots, N - 1$ and compared $\tilde{\phi}(j)$ from coset arithmetic to Grover’s $f(j) = [j = j_0]$. All values agree. **C8 REPRODUCED.**

5 Verdict

REPLICATED. Every finite/quantitative claim in the paper (C1–C8) is reproduced to machine precision by real Qiskit statevector simulation plus sympy group-theoretic computation. C9 follows from C6/C7 plus the cited Hallgren–Russell–Ta-Shma theorem. C10 (Zalka’s asymptotic $\Omega(\sqrt{N})$ bound) is out of scope for a finite numerical replication.

6 What worked / what didn't

What worked.

- Standard textbook Grover circuit at $N = 4, 8, 16$ matches $\sin^2((2k+1)\theta)$ to $\sim 10^{-14}$, so Qiskit's `Statevector` is a faithful oracle for verifying the paper's closed-form success probability.
- Direct construction of the $N \times N$ permutation unitary U_g for $g \in S_N$ and its action on the Grover statevector cleanly separates Stab_{j_0} (fidelity = 1) from generic permutations (fidelity $\sim 1/N$ or smaller). This is the strongest direct numerical evidence for the paper's "hidden symmetry" claim.
- sympy's `PermutationGroup` on S_3, S_4, S_5 made the coset arithmetic (Parts C, D, F) trivial to verify exhaustively.

What didn't (or couldn't) work.

- Marker and Nougat were not installed in this environment, so `extraction/marker.md` and `extraction/nougat.mmd` are Poppler `pdftotext` fallbacks (with header notes documenting this). This follows the same convention as the sibling `QC-quant-ph-0102014-nonabelian-hidden-sul` replication in QC-200.
- Directly simulating the "standard non-abelian QHS algorithm on S_N " as a quantum circuit is impractical for $N \geq 4$ ($|S_4| = 24$, requires 5 qubits just for the group register, plus full non-abelian Fourier transform which is not a built-in in Qiskit). We therefore verify Section 9 structurally (Parts D, E) rather than by running QHS end-to-end. This is standard: the paper's own argument is group-theoretic, not algorithmic.
- C10 (Zalka's asymptotic bound) is not numerically testable at a single N ; skipped by design.

7 Open Questions

Q1. Concrete construction of a "modified" non-abelian QHS that sees conjugacy structure. The paper leaves as its central open question whether there is a modified non-abelian QHS algorithm on S_N that solves Grover's HSP. Our Part-D verification pins the failure mode down: it is entirely explained by (i) Hallgren–Russell–Ta-Shma's normal-subgroup restriction and (ii) mutual conjugacy of the stabilisers. Any hypothetical modification must break at least one of these, e.g. by conditioning on a coset representative rather than on a Fourier label alone. *Next steps.* Implement a coset-state-based sampler (not just character-based) that measures in a basis mixing different irreducible components; test on S_4 whether such a sampler achieves $\Omega(1)$ success on Grover's HSP at $N = 4$, and compare its query count to the $O(\sqrt{N}) = 2$ Grover-optimum. Zalka's bound puts a hard ceiling at $O(\sqrt{N})$.

Q2. Noise-degradation of stabiliser invariance. Our Part-B fidelity of Stab_{j_0} -action on the Grover state is exactly 1 in the noise-free statevector. Under realistic depolarising or amplitude-damping channels the invariance should decay at a rate that is a diagnostic for how much "hidden symmetry" the physical device still respects. *Next steps.* Repeat Part B on Qiskit's density-matrix simulator with per-gate depolarising noise $p \in \{10^{-4}, 10^{-3}, 10^{-2}\}$; plot mean Stab -fidelity vs p and fit the decay. Then run on `ibm_kyoto` or a small trapped-ion device for $N = 4$.

- Q3. Smallest group evading Section-9’s obstruction.** What is the smallest finite group G with subgroup H such that (a) H is non-normal, (b) the largest normal subgroup of G in H is $\{e\}$, (c) the Stab-analog subgroups are pairwise conjugate, and yet some published non-abelian QHS algorithm still identifies H ? Any such example carves out precisely the “modified QHS” the paper asks about. *Next steps.* Survey Ivanyos–Magniez–Santha, Moore–Rockmore–Russell–Sniady, and Kuperberg (dihedral) for known efficient non-abelian QHS instances and check condition (b) for each.
- Q4. Multiple-marked-element extension.** Does the paper’s “HSP = Grover” reformulation extend cleanly to Grover with $M > 1$ marked elements? The stabiliser becomes $\text{Stab}_{\{j_1, \dots, j_M\}} = \bigcap \text{Stab}_{j_i}$, which is smaller. Does Section 7 still go through, and does Section 9’s normal-content-trivial obstruction still hold? *Next steps.* Extend `code/grover_hsp.py` with an M -marked variant (oracle = -1 on any $j \in J$) for $(N, M) \in \{(4, 2), (8, 2), (8, 3), (16, 4)\}$; re-verify $P_{\text{succ}} = \sin^2((2k+1) \arcsin \sqrt{M/N})$; re-check Section 9-(i) for setwise stabilisers.
- Q5. Standard-block-only QHS?** Part E shows $\text{Ind}_{\text{Stab}_0}^{S_N}(\mathbf{1}) = \mathbf{1} \oplus \text{std}$. If we prepared the coset state $|\text{Stab}_{j_0}\rangle$ and projected onto the standard-irrep block, does the resulting reduced density matrix depend on j_0 ? A “yes” would suggest a restricted QHS that identifies j_0 by measuring only within the standard block. *Next steps.* For $N = 4$ explicitly construct the standard-representation projector P_{std} of S_4 (dim 3) in the regular-rep basis; compute $\text{Tr}[|\text{Stab}_j\rangle\langle\text{Stab}_j| P_{\text{std}}]$ for $j = 0, \dots, 3$; check pairwise orthogonality.

References

- [1] S. J. Lomonaco, Jr. and L. H. Kauffman, *Is Grover’s Algorithm a Quantum Hidden Subgroup Algorithm?*, arXiv:quant-ph/0603140 (2006).
- [2] S. Hallgren, A. Russell, A. Ta-Shma, *The Hidden Subgroup Problem and Quantum Computation Using Group Representations*, SIAM J. Comput. 32(4), 916–934 (2003).
- [3] C. Zalka, *Grover’s quantum searching algorithm is optimal*, Phys. Rev. A 60(4), 2746–2751 (1999), arXiv:quant-ph/9711070.